Inventory Analysis Guidebook for a Retail Liquor Company

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Background

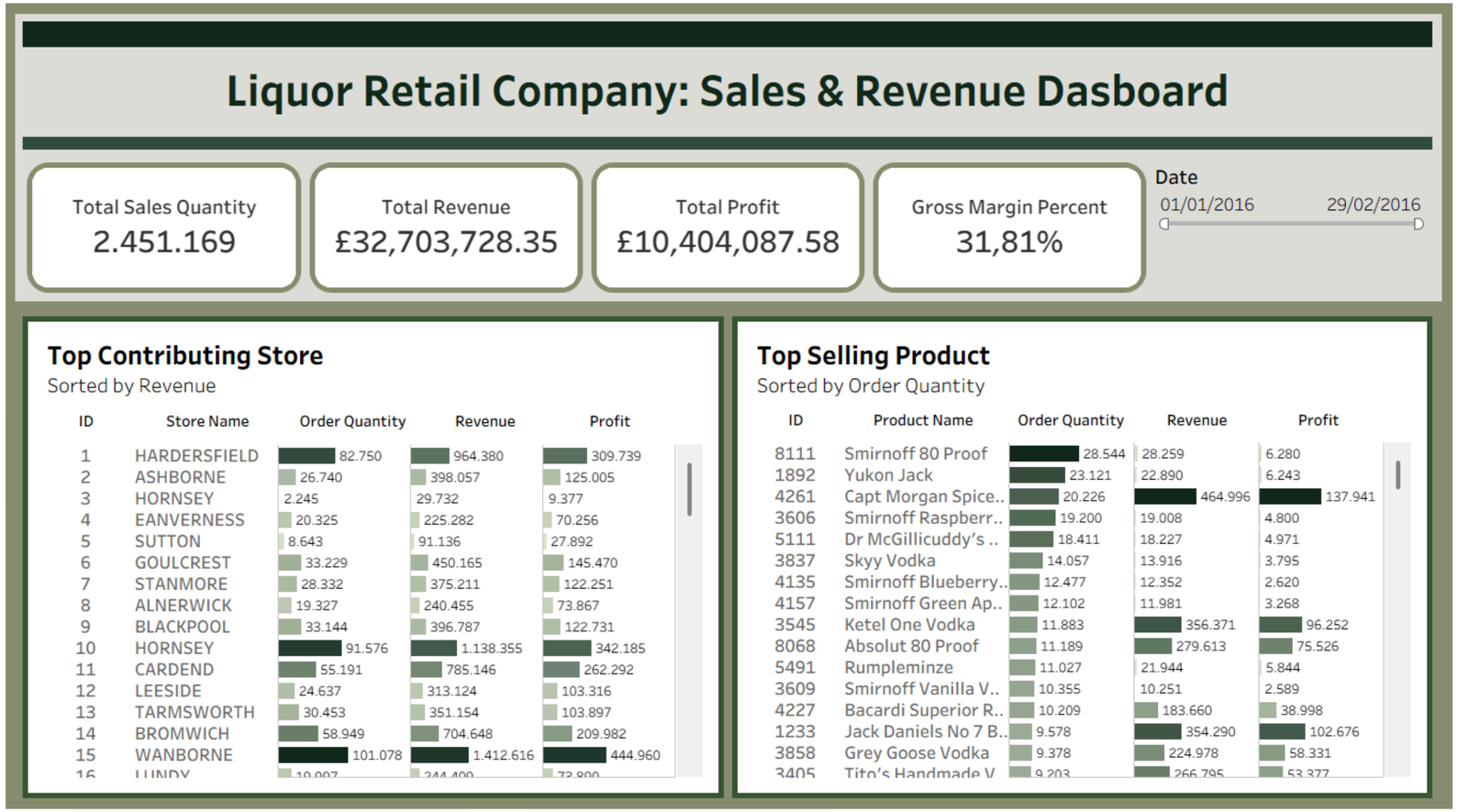
A retail company operating in the alcoholic beverage industry, offering a variety of liquor products across multiple stores. The company manages inventory at the store and product level, including different brands and sizes.

The dataset used in this analysis is sourced from

kaggle

For further details about the dataset,
visit www.kaggle.com

Dashboard



public.tableau.com/app/profile/khalila.izzatunnisa

Inventory Analysis Framework

Inventory analysis is the process of evaluating stock levels, movement, and turnover to optimize inventory, minimize carrying costs, and prevent stockouts. It enables businesses to determine the ideal amount of stock to meet demand efficiently, directly increasing profitability and cash flow.

Key Question	What to focus?	How fast it moves?	How long it last?	When to act?
Methodology/Approach	ABC Analysis	FSN Analysis Inventory Turnover Ratio	Days of Supply	Reorder Point
Objective	Prioritize items that drive the most business value	Understand how quickly inventory flows	Ensure inventory coverage aligns with demand	Determine the right timing to reorder

ABC Analysis

Definition

ABC analysis is an inventory management technique that determines the value of inventory items based on their importance to the business.

The Pareto Principle

The Pareto Principle says that most results come from only 20% of efforts or causes in any system. Based on Pareto's 80/20 rule, ABC analysis identifies the 20% of goods that deliver about 80% of the value.

Analytical Step

Calculate cumulative for revenue → Calculate the percentages for cumulative → Create ranking by percentage

Classification

Class	Importance	Total Inventory (%)	Consumption Value (%)	Controls	Records
Class A	High Value	10 - 20%	70 - 80%	Tight	High Accuracy
Class B	Medium Value	30%	15%	Medium	Good
Class C	Low Value	50%	5%	Basic	Minimal

Source: www.netsuite.com

Fast-Slow-Non (FSN) Analysis

Definition

FSN analysis is an inventory management technique which divides goods based on how fast they are used or sold and how long they stay in storage.

Parameters

Consumption Rate

The rate at which an item is consumed or expended during a specific period.

= Total units issued ÷ time period

Average Stay

The average duration that a specific item stays in the warehouse until it's sold.

= Total days items are held ÷ total items received (including opening stock)

Analytical Step

1. Calculate consumption rate and average stay
2. Calculate cumulatives for both parameters
3. Calculate the percentages for both cumulatives
4. Create ranking by percentages

Classification

Rank	Consumption Rate	Average Stay
Top 70%	Fast-Moving	Non-Moving
Middle 20%	Slow-Moving	Slow-Moving
Bottom 10%	Non-Moving	Fast-Moving

Category	Consumption Rate-Average Stay
Fast-Moving	F-F & F-S
Slow-Moving	F-N, S-F, S-S
Non-Moving	S-N, N-F, N-S, N-N

Source: www.shipbob.com

Inventory Analysis Metrics

Inventory Turnover Ratio

The amount of time that passes from the day an item is purchased by a company until it is sold.

= Cost of goods sold ÷ average inventory

Cost of Goods Sold

Cost of inventory that has been sold during a period.

Average Inventory

Average value of inventory held over a period.

= (Beginning inventory + ending inventory) / 2

Days of Supply

The number of days it would take your supply to run out if not replenished.

= Current inventory x average demand

Average Demand

Average number of units demanded per period (e.g., daily sales).

= Total demand ÷ number of periods

Reorder Point

The signal when to place a new order to avoid stock-outs.

= (Average lead time x average demand) + safety stock

Lead Time

Time between placing an order and receiving the goods.

Safety Stock

Extra inventory held to prevent stockouts due to uncertainty.

= Z-score x standar deviation of lead time

Action & Optimization

ABC x FSN Matrix: A Category

Segment	Problem	Action	Goal
Fast Moving	High demand items with high business impact → High risk of stockouts	Maintain high service level; Increase replenishment frequency; Optimize reorder point and safety stock; Improve demand forecasting accuracy	Ensure product availability and protect revenue
Slow Moving	High-value items with low movement → Capital tied up in inventory	Re-evaluate demand forecasts; Consider promotions, bundling, or markdowns; Reduce order quantities / frequency; Review assortment relevance	Reduce excess inventory and improve cash flow
Non Moving	Critical value items with no movement → Major dead stock risk	Stop replenishment immediately; Consider write-off if necessary; Liquidate through aggressive discounting; Investigate root cause	Minimize financial loss and free up working capital

Action & Optimization

ABC x FSN Matrix: B Category

Segment	Problem	Action	Goal
Fast Moving	Moderate value but consistent demand ↓ Risk of inefficiency if overcontrolled	Maintain balanced inventory policy Apply standard replenishment rules Monitor trends periodically	Achieve efficiency with controlled availability
Slow Moving	Moderate value with low turnover ↓ Potential overstock risk	Adjust order quantities Monitor demand variability Avoid overstocking	Maintain optimal inventory levels
Non Moving	Low movement with moderate value ↓ Hidden dead stock	Limit or stop replenishment Run targeted promotions Review product necessity	Prevent buildup of inactive inventory

Action & Optimization

ABC x FSN Matrix: C Category

Segment	Problem	Action	Goal
Fast Moving	Low-value but high volume items → Operational importance but low financial risk	Simplify replenishment (bulk ordering); Maintain sufficient stock levels; Automate control where possible	Ensure operational continuity at low cost
Slow Moving	Low value and slow movement → Inefficient space usage	Reduce stock levels; Order only when necessary; Consider SKU rationalization	Optimize storage and reduce handling cost
Non Moving	Low value, no movement → Pure dead stock	Eliminate or liquidate inventory; Stop reordering permanently	Clean up inventory and reduce holding cost

Analysis Code

The analysis is performed using

colab

Access and run the analysis code at
www.colab.research.google.com

GitHub Repository

https://github.com/khalilaizztnns/